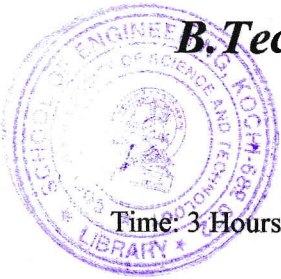


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Time: 3 Hours

B.Tech. Degree IV Semester Examination April 2019**CE 1402 (A/B) SURVEYING II**
(2012 Scheme)

Maximum Marks: 100

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) List the elements of a simple curve and give the expression for finding each.
- (b) What are the types of vertical curves?
- (c) What is a satellite station? Also define the terms eccentric distance and reduction to centre.
- (d) What are the rules for figure adjustment of a plane triangle?
- (e) Define the terms spherical excess, celestial sphere, zenith and nadir.
- (f) Express $4^{\text{h}}34^{\text{m}}13^{\text{s}}$ in degrees, minutes and seconds.
- (g) List the purposes for sounding.
- (h) Write the principle underlying the method of Terrestrial photogrammetry.

PART B

(4 × 15 = 60)

- II. (a) Two roads having a deviation angle of $45^{\circ}48'$ are to be joined by a 180 m radius curve. Chainage of intersection of tangents is 3123.8 m. Calculate the necessary data if the curve is to be set by offsets from chords produced. Assume 30 m chain. (10)
- (b) Write the advantages of providing super-elevation. (5)

OR

- III. (a) Two straights AB & BC are intercepted by a line MN. The angles AMN and MNC are 145° and 140° respectively. Radius of first arc is 400 m and that of second arc is 600 m. Find the chainages of tangent points and point of compound curvature. The chainage of point of intersection is 5555 m. (10)
- (b) How do you determine the length of transition curve? (5)

- IV. (a) Briefly explain the classification of triangulation. (7)
- (b) There are 2 stations A and B at elevations of 200 m and 1000 m, respectively. The distance between A and B is 100 km. If the elevation of a peak P at a distance of 40 km from A is 300 m, show that stations A and B are intervisible. (8)

OR

- V. (a) State the laws of weight. (7)
- (b) Three angles A, B and C of a triangle ABC are $49^{\circ}20'13.4''$, $50^{\circ}29'15.2''$ and $80^{\circ}10'25.4''$ with weights 1, 2 and 3 respectively. Find the Most Probable Values of A, B and C by the theory of least squares. (8)

(P.T.O.)

- VI. (a) Explain Astronomical Triangle with a sketch. (5)
 (b) Find GMT corresponding to the following LMT: (10)
 (i) $9^h40^m12^s$ A.M. at a place in longitude $42^\circ36'W$
 (ii) $4^h32^m10^s$ A.M. at a place in longitude $56^\circ32'E$

OR

- VII. (a) Explain the horizon system of co-ordinates. (5)
 (b) Determine the hour angle and declination of a star from the following data: (10)
 Altitude of star = $22^\circ36'$
 Azimuth of the star = $42^\circ W$
 Latitude of the place of observation = $40^\circ N$

- VIII. (a) Explain the method of locating soundings by cross rope. (5)
 (b) The following observations were made to 3 stations A, B and C from a boat at O with the help of a sextant. Station B and O are on the same side of AC. Distances $AB = 4000$ m & $BC = 4995$ m. Find the distance of boat from the three stations if, $\angle AOB = 30^\circ25'$, $\angle BOC = 45^\circ25'$, $\angle ABC = 130^\circ10'$. (10)

OR

- IX. (a) Write the principle of stereoscopic fusion. (7)
 (b) Determine the number of photographs required to cover an area of $8 \text{ km} \times 12.5 \text{ km}$, if the longitudinal lap is 60% and the side lap is 30%. The scale of an aerial photograph is $1 \text{ cm} = 100 \text{ m}$. The photograph size is $20 \text{ cm} \times 20 \text{ cm}$. (8)
